

SCC.453: Natural Language Processing Research Project

NFL Commentary: A Study of Temporal Language Drift in Commentary using Lexical Shift Analysis and Classification Models

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Link to Kaggle Dataset: [American Football Dataset](#)

Abstract

This report details the data, methodology, results, and meaning of the findings regarding the language shift in NFL broadcast commentary, spanning games from 1970 to 2019. The Football Corpus (Merullo et al., 2019) was the foundation for the data, which provided 167,340 player mentions after preprocessing, cleaning and token blacklisting had been performed. The key techniques utilised for this research project consisted of bootstrapped log-ratio shift analysis, rolling lexical divergence, and within-and cross-decade evaluations, paired with McNemar's statistical test. We applied this analysis to 5 key player positions, where all of them showed statistically reliable lexical shifts. The position Wide Receiver (WR) exhibited the greatest divergence with a cosine distance of 0.257 between the 1970 and 2015 5-year bins. Within-decade classifiers obtain Macro-F1 scores of 0.31 to 0.38, five times above the baseline. Cross-decade evaluation collapses to an F1 score ≈ 0.28 to 0.33. To emphasise this, McNemar's test shows a statistically significant difference between within-decade and cross-decade metrics when directly comparing the 1970s and 2010s. When viewing this month by month we observe a drift in language that becomes a shift when viewing from the extreme points in time.

1 Introduction

American football broadcasts represent one of the most temporally deep annotated spoken corpora in existence. The dataset spans more than half a century of recorded play-by-play commentary. Language change over time is well-documented in written corpora (Hamilton et al., 2016), but spoken broadcast commentary presents a distinct case: it is intertwined with the tactical evolution of the sport, and shifts in broadcasting that are due to social changes and technological advances.

The broader question we are examining can be broken down into two distinct, yet connected parts.

Question 1: Is there a shift in the language used to describe NFL players between 1970 and 2019, and does this vary based on player position?

Question 2: Does temporal shift impair the ability of classification models when trained and tested on different decades?

The techniques we are implementing to answer both the overall question and the two parts outlined above are: Bootstrap Log-Ratio Shift Analysis with 95% Confidence Intervals across five player positions, Rolling Lexical Divergence with the creation of 5-year bins to understand within-decade changes, wrapping it up with a McNemar significance test to assess cross-decade changes.

2 Related Work

Hamilton et al. (2016) show that word embeddings trained on different time periods reveal systematic semantic drift, with some word classes shifting faster than others. The log-ratio shift approach we use is related to methods in computational sociolinguistics (Eisenstein et al., 2014) for measuring vocabulary divergence between communities over time. Within sports NLP, Merullo et al. (2019) introduce the football corpus (used in this project) to study racial bias in commentary, showing that descriptions of white and non-white players differ systematically. Our work is complementary: where they examine *what* is said about players, we examine *how that language has evolved*. On transfer learning, Blitzer et al. (2006) formalise domain adaptation as a problem of covariate shift; temporal shift presents the same challenge but continuously and without explicit domain boundaries.

3 The Data

We use the [Football Dataset](#) (Merullo et al., 2019), comprising 1,455 full NFL and NCAA game broadcasts from 1960 to 2019. Each record is a window of tokens surrounding a tagged player mention with metadata including position and year. We have restricted the data to just include NFL games and five positions, a mix of both offensive and defensive positions, allowing for vague insight into understanding differences in language on a more vast level. The offensive positions: quarterback (QB), running back (RB), and wide receiver (WR). The defensive positions: defensive back (DB), and linebacker (LB), retaining mentions from 1970 onwards. This yields 167,340 mention windows (Table 1).

Decade	DB	LB	QB	RB	WR
1970s	1,901	1,676	3,160	3,498	2,531
1980s	2,345	1,629	5,814	3,439	3,419
1990s	4,116	2,132	6,008	3,789	4,467
2000s	9,285	3,641	21,167	11,286	10,581
2010s	11,537	5,329	17,707	12,388	14,495

Table 1: Mention counts by decade and position.

A key preprocessing challenge is the name token blacklist. By removing every token appearing in player or team names strips common English words that double as surnames (*hill, young, long*). Therefore, it is paramount to cross-reference name tokens against a curated common-word set before blacklisting, providing a set of 10,954 name-specific tokens. Some limitations of the data, which we can try to minimise: imbalance of classes, more data points for the 2000s and 2010s opposed to 1960s (which we have removed due to sparsity), the linguistic variation between *proper English* and *sports commentary*, and the dataset using short commentary windows around player mentions instead of full sentences or play-by-play descriptions.

4 Methodology

Preprocessing Text is lowercased, HTML tags removed, and tokens under two characters removed. A custom stop-word list is installed to extend scikit-learn’s English stop words with *generic football* terms (*game, yard, ball*) that transcend positions and decades. Finally, the <player> tag is removed before analysis.

Lexical shift We will compute log-ratio shift between our refined 1970s and 2010s data.

Determining our shift score can be done by calculating $\log(\frac{p_b(t)}{p_a(t)})$, where t denotes the term and $p_a(t)/p_b(t)$ denotes the normalised frequency for the term in corpus "a" and "b". In order to establish reliability, we utilise the aforementioned approach of bootstrapping with 95% confidence intervals. Resampling with replacement being performed 100 times, with only terms that obtained confidence intervals not crossing 0 being taken. On top of this, the features used are unigrams and bigrams with a threshold document frequency of 25.

Lexical divergence To delve deeper into change over time, segments of 5-year spans were created opposed to the default 10 years. Our segments will then be shown as follows: 1970, 1975, ..., 2015. The years will go into the bin rounded down to the nearest 5, for example: 1974 goes to 1970 and 2016 goes to 2015. Now that we’ve established our windows, we compute pairwise cosine similarity scores between TF-IDF vectors for the respective aggregated time periods, granting 225 pair distances per position. The final distance is mathematically denoted as: $d(S_i, S_j) = 1 - \cos(\mathbf{v}_i, \mathbf{v}_j)$ with S_i being the segment for time period, i , and $\mathbf{v}_i$ being its respective TF-IDF vector.

Classification For this task, which is more associated with our second part of the research question, we want to select 3 models to perform classification. This way, we can make sure the results aren’t model-dependent. Ultimately deciding on Multinomial Naive Bayes (NB), Logistic Regression (LR), and LinearSVC (SVC). Training each model on TF-IDF features, consistently using unigrams and bigrams, min_df = 2, and a maximum number of features being 5000. Within-decade performance uses five-fold stratified CV with 300 balanced examples per class. Cross-decade transfer trains on one decade and tests on another (150 examples/class). Additionally, setting a baseline provides a lower bound, which should be identical due to building a balanced sample which contains an equal number of each class in the set.

Significance testing McNemar’s test (McNemar, 1947) compares within-decade and cross-decade classifiers on a shared 2010s test set, using three

disjoint splits to avoid leakage. The continuity-corrected statistic is applied to matched correctness vectors. Revealing if the assumed drop in performance between within-decade and cross-decade is actually statistically significant. The McNemar’s Chi-Squared is calculated as follows: $\chi^2 = \frac{(|b-c|-1)^2}{(b+c)}$ where b denotes where the within-decade model is correct and the cross-decade is incorrect, c is the opposite of b .

5 Results

Lexical shift Table 2 highlights the top statistically reliable shifts for both increasingly and decreasingly used terms. The 2010s is encapsulated with the rise of tactical vocabulary (*route*, *pocket*, *coverage*, *sack*, *target*) and more expressive language (*huge*, *nice*, *guys*). The 1970s retain generic positional references (*men*, *linebackers*, *wide receivers*) and hints of older broadcast style (*number number*, indicating jersey-number identification).

More 2010s		More 1970s	
Term	LogR	Term	LogR
route	+2.92	replacing	−2.32
huge	+2.38	number number	−2.18
special	+2.15	men	−2.12
nice	+1.79	bench	−2.00
pocket	+1.66	pattern	−1.88
sack	+1.50	linebackers	−1.84
coverage	+1.38	number	−1.56
target	+1.37	wide receivers	−1.39

Table 2: Top 8 bootstrapped log-ratio (LogR) shifts (1970s vs. 2010s) for all positions

Position-specific shifts reveal further structure. WR commentary sees *flanker* and *flank* becoming obsolete terms that are replaced by *route*, *matchup*, and *snap*. DB commentary moves from general terms such as *safety man* or *tight safety* towards schematic terminologies of *nickel* and *coordinator*. QB commentary transitions from jersey-number references towards evaluative language: *starter*, *seeing*, and *accurate*.

Lexical divergence Table 3 shows mean cosine distance grows monotonically for all the time windows and selected positions, confirming continuous drift rather than abrupt shift. Notably, WR exhibits the greatest rolling divergence compared to the remaining evaluated positions, peaking at a distance of 0.257 at the 45 year interval (1970 to 2015 bins). This indicates consistency with the phasing out of *flanker*-esque terminology. QB is the most stable (0.133), plausibly because "passing-game" vocab-

ulary has remained central throughout. This train of thought can also be applied to RB, where they predominantly run the ball and terminology revolving around it hasn’t become as expressive in the current era opposed to the 1970s. LB and DB show intermediate but substantial divergence.

Pos.	5yr	15yr	25yr	35yr	45yr
WR	0.050	0.091	0.140	0.212	0.257
LB	0.086	0.112	0.127	0.172	0.222
DB	0.070	0.088	0.117	0.153	0.206
RB	0.045	0.062	0.088	0.114	0.149
QB	0.038	0.058	0.079	0.108	0.133

Table 3: Mean cosine distance between 5-year windows at selected temporal gaps, by position.

Classification For the within-decade approach, Table 4 illustrates Macro-F1 scores. The baseline score is included to enable a comparison, which resoundingly demonstrates that each model selected vastly outperforms it when predicting class (QB or WR, for example). Confirming that there are elements in commentary language for predicting positions. Across the decades tested, Logistic Regression outperforms both Naive Bayes and LinearSVC. This could be attributed to its linear decision boundaries generalising more effectively, with a balanced class sample set, than NB with conditional independence assumptions or the hard margins imposed by SVC. The scores all decrease over this time frame, which potentially reflects somewhat of a convergence in terminology.

Decade	NB	LR	SVC
1970s	0.371	0.383	0.369
1990s	0.320	0.336	0.317
2010s	0.310	0.317	0.289
Baseline		0.067	

Table 4: Within-decade Macro-F1 (5-fold CV, 300 per class)

Having evaluated the within-decade approach, Table 5 outlines cross-decade for LR, specifically. The diagonal values represent within-decade without the 5-fold cross-validation applied. There is a dramatic collapse in the F1 scores once the training and testing sets don’t align on a decade. Training on a more recent decade and testing on an older one is more effective (although scores remain weak) than the inverse, ruling out the explanation that richer vocabulary is expressed more in recent broadcasts.

Train → Test	1970s	1990s	2010s
1970s	0.829	0.302	0.282
1990s	0.326	0.829	0.313
2010s	0.329	0.327	0.804

Table 5: Cross-decade Macro-F1 (LR: 150 perclass). Bold = within-decade (no CV).

Significance testing To stay consistent with the prior evaluation, we will look predominantly at LR for this (although all 3 approaches perform almost equally on this test and reflect the same outcome, as indicated in Table 6). It is evident that based on McNemar’s test, due to our $p\text{-value} < 0.001$ for all classifiers, there is a significant drop in performance between within-decade and cross-decade when strictly comparing the 1970s and 2010s.

Metric	NB	LR	SVC
Within Correct	439	460	516
Cross Correct	24	21	2
McNemar (χ^2)	370.1857	398.8441	508.0483
p-value	0.000	0.000	0.000
Significance	TRUE	TRUE	TRUE

Table 6: McNemar Significance Test: Within-decade vs Cross-decade predictions across classifiers

6 Conclusions, Limitations, and Future Work

This research demonstrates that NFL broadcast commentary is not a static linguistic domain but a rapidly evolving one. Through bootstrapped log-ratio analysis, we identified a definitive shift from generic descriptors and jersey-number identification in the 1970s toward highly technical, schematic vocabulary in the 2010s. The WR position serves as the epicentre of this evolution, exhibiting a lexical divergence score of 0.257, nearly double that of the QB position, achieving a score of 0.133, likely reflecting the transition of in-game tactics from the 1970s to modern day.

Our classification experiments confirmed that this drift has tangible consequences for machine learning. We observed a significant "temporal transfer collapse"; models trained on historical data are essentially "blind" to modern positional contexts. McNemar’s test yielded Chi-Squared scores as high as 508.05 ($p < 0.001$), definitively rejecting the null hypothesis that within-decade and cross-decade models perform equally. This suggests that in sports NLP, a model’s "shelf life" is remarkably short, and longitudinal studies must account for the fact that

a "Wide Receiver" in 1970 is linguistically distinct from a "Wide Receiver" in 2019.

Limitations Despite the robustness of the statistical tests, this study is limited by predominantly looking at decades, as opposed to using the exact year, which could give further insight at the cost of increased runtime and not as generalised as a set time period of a decade or 5 years. Furthermore, while the bootstrap resampling ($N=100$) ensures reliability, the class imbalance in the 1970s (where data is significantly more sparse) may still introduce some variance in the stability of the oldest embeddings. For reproducibility purposes, consistent random seeds were selected, which improves reproducibility but slightly reduces robustness, and the results could be biased towards the set data split. Finally, this approach tracks lexical change (word choice) but does not fully model semantic change (words that stay the same but change meaning).

Future Work Future research should move beyond count-based methods to Diachronic Word Embeddings (e.g., TWEC or Procrustes-aligned Word2Vec) to map how words like "target" or "slot" have shifted their geometric positions in semantic space.

Appendix

A Stop Words

```
CUSTOM_STOP = {
    "just", "good", "got", "going", "game", "play", "time",
    "yard", "yards", "line", "ball", "second", "right", "left",
    "first", "one", "two", "three", "four", "five", "six",
    "seven", "eight", "nine", "ten", "back", "down", "really",
    "little", "lot", "look", "looking", "come", "comes", "say",
    "said", "player",
}

ANALYSIS_STOP = set(sk_text.ENGLISH_STOP_WORDS) | CUSTOM_STOP
```

B Mentions per Decade

decade	DB	LB	QB	RB	WR	
1970	1901	1676	3160	3498	2531	
1980	2345	1629	5814	3439	3419	
1990	4116	2132	6008	3789	4467	
2000	9285	3641	21167	11286	10581	
2010	11537	5329	17707	12388	14495	

C Model / Classifier Parameters

```
def models():
    return {
        "NaiveBayes": Pipeline([
            ("tfidf", TfidfVectorizer(
                stop_words = "english", ngram_range = (1, 2),
                min_df = 2, max_features = 5000)),
            ("clf", MultinomialNB()),
        ]),
        "LogisticRegression": Pipeline([
            ("tfidf", TfidfVectorizer(
                stop_words = "english", ngram_range = (1, 2),
                min_df = 2, max_features = 5000)),
            ("clf", LogisticRegression(
                max_iter = 1000, C = 1.0, solver = "lbfgs", random_state = 42)),
        ]),
        "LinearSVC": Pipeline([
            ("tfidf", TfidfVectorizer(
                stop_words = "english", ngram_range = (1, 2),
                min_df = 2, max_features = 5000)),
            ("clf", LinearSVC(max_iter = 2000, C = 1.0, random_state = 42)),
        ]),
    }
```

D Bootstrapping Shift (csv file)

term_incre	lr_increasi	ci_lo_incre	ci_hi_incre	Significanc	term_decr	lr_decreas	ci_lo_decr	ci_hi_decr	Significanc
route	2.924517	2.316338	4.109827	TRUE	replacing	-2.32251	-3.60022	-1.45783	TRUE
huge	2.377974	2.488465	3.521363	TRUE	number nu	-2.17941	-3.37948	-1.50282	TRUE
special	2.149132	2.549935	3.525991	TRUE	men	-2.12406	-3.43472	-1.49762	TRUE
nice	1.792457	1.382258	2.557144	TRUE	bench	-2.00092	-3.45689	-1.4803	TRUE
guys	1.714679	1.332860	2.238476	TRUE	pattern	-1.87622	-3.61983	-1.08931	TRUE
pocket	1.663624	1.112101	2.438657	TRUE	linebacker	-1.84153	-2.45928	-1.39619	TRUE
drive	1.520523	1.030393	2.085002	TRUE	youll	-1.83003	-3.10149	-1.23951	TRUE
sack	1.496426	0.809301	2.424211	TRUE	thousand	-1.74269	-2.90867	-0.98475	TRUE
numbers	1.484156	0.829143	2.623081	TRUE	middle line	-1.6939	-2.61989	-1.11357	TRUE
snap	1.468159	0.835367	2.535024	TRUE	number	-1.55768	-1.75678	-1.39093	TRUE
nfl	1.433512	0.990861	1.833993	TRUE	ill tell	-1.44131	-2.34459	-0.7901	TRUE
coverage	1.378207	1.065722	1.746240	TRUE	wide receiv	-1.39297	-2.02359	-0.96987	TRUE
target	1.366373	0.838576	2.132921	TRUE	motion	-1.32213	-1.77766	-0.99335	TRUE
mean	1.348354	0.878901	2.192345	TRUE	draw	-1.30085	-2.19461	-0.67323	TRUE
underneat	1.346955	0.897736	1.885846	TRUE	running ba	-1.29038	-1.9267	-0.74999	TRUE
round	1.343383	0.711667	2.193962	TRUE	itll	-1.27268	-1.73334	-0.87246	TRUE
injury	1.301834	0.592006	2.193246	TRUE	lineup	-1.27268	-2.18303	-0.58692	TRUE
opportunit	1.295755	0.815109	2.004919	TRUE	wide	-1.23437	-1.4944	-1.02471	TRUE
thanks	1.292265	0.781516	2.289489	TRUE	loose	-1.22389	-1.91579	-0.67489	TRUE
possession	1.248590	0.544233	1.916724	TRUE	finally	-1.19404	-1.73571	-0.65771	TRUE
season	1.239959	0.875356	1.578807	TRUE	white	-1.18067	-1.79207	-0.75933	TRUE
talked	1.232841	0.738207	1.802437	TRUE	covered	-1.11853	-1.51645	-0.71859	TRUE
catch	1.175318	0.865321	1.473823	TRUE	cornerbac	-1.09036	-1.69811	-0.66301	TRUE
touchdown	1.139315	0.559269	1.763535	TRUE	crowd	-1.06974	-1.653	-0.66062	TRUE
teams	1.102221	0.647791	1.805965	TRUE	backs	-1.06852	-1.48402	-0.78567	TRUE

Figure 1: Overall

more_201	lr_up	Confidenc	Confidenc	more_197	lr_down	Confidenc	Confidence	Interval	High
fourman	2.911422	2.535938	3.181587	4number nu	-3.54692	-3.9534	-3.03504		
seeing	2.722180	2.253417	3.03499	jj	-3.52393	-4.40343	-2.73921		
nice	2.688279	1.165884	3.709054	setbacks	-3.22846	-3.60986	-2.70252		
nice job	2.679621	0.179161	3.160423	62youngster	-3.02082	-3.5034	-2.52094		
starter	2.635169	3.2.180372	0.950462	22triggers	-2.85377	-3.36426	-2.06402		
opening dr	2.434498	1.898361	5.2.812871	6cheering	-2.47078	-3.48499	-1.50409		
gun	2.377340	2.1.940620	1.721373	6ill tell	-2.4483	-3.6241	-1.59372		
accurate	2.316715	1.749925	2.645770	1emotion	-2.40178	-3.3416	-1.59725		
comeback	2.252177	0.173695	1.2.552551	0ed	-2.32768	-3.34098	-1.25727		
route	2.252177	0.149214	65.3.444323	6following	-2.32768	-3.33544	-1.54522		
shoulder ir	2.252177	0.167502	2.2.616434	8evening	-2.24763	-3.20015	-1.4178		
eyes	2.229704	2.1.264272	3.473924	1flag goes	-2.24763	-3.33921	-1.45388		
clearly	2.183184	1.1.344962	4.2.624459	0overthrow	-2.16062	-3.28388	-1.1532		
fourman ru	2.183184	1.1.549855	4.2.661555	7ram	-2.16062	-3.36289	-1.35053		
issue	2.183184	1.1.620713	2.6.631241	0screen pat	-2.16062	-3.32657	-1.15419		

Figure 2: QB

more_201	lr_up	Confidenc	Confidenc	more_197	lr_down	Confidenc	Confidence	Interval	High
nice job	2.813030	7.2.403767	7.3.098774	7wide wide	-3.94157	-4.26939	-3.39426		
huge	2.782259	2.407011	5.3.068009	6flanker	-3.77452	-4.10751	-3.38744		
matchup	2.612360	0.2.247388	3.2.894569	9goes wide	-3.45607	-3.83189	-2.8794		
route	2.612360	0.1.977965	4.0.223721	1flank	-3.24843	-3.72559	-2.55078		
snap	2.494576	1.980882	2.808040	4goulash	-3.24843	-3.73273	-2.54538		
including	2.452017	3.1.955863	2.8.831258	3grambling	-3.24843	-3.7166	-2.76964		
round pick	2.407565	2.0.541396	2.7.761458	0setback	-3.24843	-3.70383	-2.53552		
youre gonr	2.407565	1.965041	2.8.834618	1deep men	-3.16838	-3.59544	-2.53656		
opportunit	2.361045	1.417455	3.2.241226	0moral	-3.16838	-3.5996	-2.46637		
weapons	2.361045	1.976966	1.2.666752	1morel	-3.16838	-3.64803	-2.55309		
practice	2.312255	4.1.703357	5.2.728000	2morale	-3.12582	-3.99286	-2.41121		
seeing	2.260962	1.1.673394	2.6.663099	9cosell	-3.08137	-3.55305	-2.47171		
underneat	2.260962	1.1.457011	2.3.464418	1number wi	-2.76292	-3.80216	-1.72208		
talked	2.220688	1.1.556015	3.3.555447	7replacing	-2.69838	-3.73051	-1.81678		
targeted	2.208694	9.1.754800	4.2.624808	6number nu	-2.62939	-3.88494	-1.73528		

Figure 3: WR

more_201	lr_up	Confidenc	Confidenc	more_197	lr_down	Confidenc	Confidence	Interval	High
weve seen	2.561587	9.1.902005	3.4.599043	0setbacks	-3.99377	-4.2738	-3.66628		
arm	2.534188	2.0.085305	3.2.912296	0seaman	-3.61428	-3.9357	-3.25932		
punch	2.534188	2.0.092597	1.2.886071	1firing	-3.1894	-3.5539	-2.65391		
seeing	2.534188	1.953150	4.2.883945	0thousand	-3.12877	-3.51052	-2.59603		
drafted	2.477030	5.1.842161	1.2.942651	8period	-3.09702	-3.46639	-2.23967		
excited	2.416405	1.589728	2.8.824585	0hildenburg	-3.06423	-3.4868	-2.42889		
sunday	2.416405	1.8.846482	2.2.743899	5remain	-3.06423	-3.49454	-2.48978		
combinati	2.351867	3.1.832819	2.2.777313	0flanker	-2.99524	-3.43877	-2.42273		
inactive	2.351867	3.1.727059	2.2.752652	0gets itll	-2.99524	-3.38018	-2.4326		
initially	2.351867	1.8.835321	2.2.739551	1number ru	-2.92113	-3.29077	-2.27767		
roster	2.282874	4.1.602933	4.2.723335	0wide wide	-2.92113	-3.40724	-2.29171		
talked	2.268486	1.5.264983	3.3.302351	3enabled	-2.84109	-3.24031	-2.15288		
challenge	2.208766	5.1.513253	7.2.626915	0enabled sf	-2.84109	-3.24031	-2.15288		
driscoll	2.208766	5.1.707780	9.2.638013	5husties	-2.84109	-3.27631	-2.21928		
gruden	2.208766	5.1.437752	3.2.655906	4quarterbar	-2.84109	-3.25468	-2.21372		

Figure 4: RB

more_201	lr_up	Confidenc	Confidenc	more_197	lr_down	Confidenc	Confidence	Interval	High
nickel	2.631562	2.3.090003	2.9.442803	0heyman	-4.88378	-5.20062	-4.50824		
coordinat	2.449241	2.1.994343	4.2.788035	8haman	-4.17083	-4.49246	-3.58155		
starts	2.449241	2.1.882845	7.2.725135	3safety mar	-3.84233	-4.18539	-3.43297		
nice job	2.343880	7.1.911485	3.2.642410	0tight safet	-3.78517	-4.18243	-3.25634		
youre gonr	2.266919	7.1.874556	2.2.594981	6number sa	-3.72454	-4.09015	-3.14033		
grab	2.226097	7.1.711474	2.2.522254	8number co	-3.66001	-4.03436	-3.1593		
match	2.226097	7.1.738947	0.2.658814	5cornerbac	-3.59101	-3.91983	-2.96272		
defensive	2.139086	3.1.632977	7.2.520227	7dropping d	-3.34985	-3.78133	-2.64221		
target	2.139086	3.1.307815	3.0.040668	6number nu	-3.34985	-4.59289	-2.51772		
tomlin	2.139086	3.1.611857	3.2.489785	3safety nur	-2.98884	-4.28444	-2.1531		
terrific	2.043776	1.1.481805	2.2.362628	0minutes se	-2.82376	-3.8129	-1.6249		
bradbury	1.992482	2.1.395068	4.2.360858	5dropping	-2.33825	-3.75858	-1.43816		
rutgers	1.992482	2.1.171936	4.2.485324	0pattern	-2.27371	-3.3301	-1.26352		
matchup	1.965814	1.1.136227	2.2.908947	6complainir	-2.25124	-3.48504	-0.94476		
opportunit	1.938415	6.1.165056	7.2.927106	1deep man	-2.15593	-3.62432	-1.04725		

Figure 5: DB

E Rolling Lexical Divergence

more_2011(lr_up	Confidenc	Confidenc	more_1971(lr_down	Confidenc	Confidence	Interval_High
nice job	2.7731716	2.5289894	3.0176021	marinero	-4.0512	-4.38087 -3.44835
guys	2.2647585	1.7812758	3.0193303	ed	-3.54038	-4.06773 -2.84998
agent	2.2403670	1.8735860	2.5988983	hacksaw	-3.54038	-3.9873 -2.99014
foul	2.2403670	1.7768718	2.5711075	hollywood	-3.54038	-3.97664 -2.75897
instead	2.2026267	1.7092616	2.5437390	number nu	-3.00138	-4.17802 -2.36433
fiveyard	2.1634060	1.6850530	2.5484012	number mi	-3.00138	-3.93961 -2.0284
terrific	2.1634060	1.5876852	2.5125972	period	-2.92727	-3.94885 -1.94039
starting	2.1432035	1.3043012	2.9955158	pinching	-2.76022	-3.7723 -1.53158
finds	2.1225840	1.6051392	2.5194632	steeler	-2.72945	-4.0032 -1.83449
concussio	2.0355726	1.5782157	2.4057585	caused	-2.66491	-3.79684 -1.57124
matchup	2.0355726	1.6489947	2.4071371	hesitation	-2.66491	-3.76448 -1.75262
points	2.0355726	1.6149011	2.3769993	michigan s	-2.66491	-3.76306 -1.70325
route	2.0355726	1.5014679	2.3627265	trees	-2.66491	-3.66596 -1.89721
special	2.0048010	1.3458895	3.2362146	marla	-2.59591	-4.00157 -1.68277
face	1.9649550	1.0219795	2.9220766	number lin	-2.44176	-3.69351 -1.69527

Figure 6: LB

Positions	Early_Time	Late_Time	Time_Gap	Cosine_Sir	Distance
DB	1970	1975	5	0.818273	0.181727
DB	1970	1980	10	0.8730142	0.12698577566176084
DB	1970	1985	15	0.8479569	0.15204300207916677
DB	1970	1990	20	0.8441926	0.15580736656376482
DB	1970	1995	25	0.8019431	0.1980568015327997
DB	1970	2000	30	0.8163995	0.1836004566098367
DB	1970	2005	35	0.8160298	0.1839701416873687
DB	1970	2010	40	0.7979705	0.20202942775340327
DB	1970	2015	45	0.7943364	0.20566355602058284
DB	1975	1980	5	0.8607151	0.13928484945629083
DB	1975	1985	10	0.8429172	0.15708275055658372
DB	1975	1990	15	0.8529986	0.14700132999006787
DB	1975	1995	20	0.8217861	0.17821384085954894
DB	1975	2000	25	0.8494617	0.1505382772024202
DB	1975	2005	30	0.8529818	0.14701810108610458
DB	1975	2010	35	0.8372440	0.16275594212192224
DB	1975	2015	40	0.8355519	0.1644480087075435
DB	1980	1985	5	0.9222334	0.077767
DB	1980	1990	10	0.9361138	0.063886
DB	1980	1995	15	0.9060061	0.093994
DB	1980	2000	20	0.9224403	0.07756
DB	1980	2005	25	0.9190623	0.080938
DB	1980	2010	30	0.9001296	0.09987
DB	1980	2015	35	0.8868250	0.1131749059858892
DB	1985	1990	5	0.9456155	0.054384428625389836
DB	1985	1995	10	0.9153966	0.084603
DB	1985	2000	15	0.928898	0.071102
DB	1985	2005	20	0.9265745	0.073425
DB	1985	2010	25	0.9110993	0.088901
DB	1985	2015	30	0.8930432	0.10695670515185729
DB	1990	1995	5	0.9513088	0.048691
DB	1990	2000	10	0.9568930	0.043106929166277785
DB	1990	2005	15	0.955871	0.044129463994881046
DB	1990	2010	20	0.9413776	0.058622
DB	1990	2015	25	0.9336003	0.0664
DB	1995	2000	5	0.9510587	0.048941271139609555
DB	1995	2005	10	0.9512219	0.048778
DB	1995	2010	15	0.9455620	0.054438
DB	1995	2015	20	0.9358504	0.06415
DB	2000	2005	5	0.973767	0.026232863542118956
DB	2000	2010	10	0.9632513	0.036749
DB	2000	2015	15	0.9488486	0.051151
DB	2005	2010	5	0.978391	0.021608846474177046
DB	2005	2015	10	0.9677298	0.032270110199692725
DB	2010	2015	5	0.970257	0.029743

Figure 7: 5-Year Windows (DB)

F Classification

Positions	Early_Time	Late_Time	Time_Gap	Cosine_Sir	Distance
DB	1970	1980	10	0.8730142	0.12698577566176084
DB	1970	1990	20	0.8441926	0.15580736656376482
DB	1970	2000	30	0.8163995	0.1836004566098367
DB	1970	2010	40	0.7979705	0.20202942775340327
DB	1975	1985	10	0.8429172	0.15708275055658372
DB	1975	1995	20	0.8217861	0.17821384085954894
DB	1975	2005	30	0.8529818	0.14701810108610458
DB	1975	2015	40	0.8355519	0.1644480087075435
DB	1980	1990	10	0.9361138	0.063886
DB	1980	2000	20	0.9224403	0.07756
DB	1980	2010	30	0.9001296	0.09987
DB	1985	1995	10	0.9153966	0.084603
DB	1985	2005	20	0.9265745	0.073425
DB	1985	2015	30	0.8930432	0.10695670515185729
DB	1990	2000	10	0.9568930	0.043106929166277785
DB	1990	2010	20	0.9413776	0.058622
DB	1995	2005	10	0.9512219	0.048778
DB	1995	2015	20	0.9358504	0.06415
DB	2000	2010	10	0.9632513	0.036749
DB	2005	2015	10	0.9677298	0.032270110199692725

Figure 8: Decade (DB)

Decade	Classifier	Accuracy	Macro_F1
1970	NaiveBaye	0.3713333	0.37056472672067364
1970	LogisticRe	0.384	0.3834334173333864
1970	LinearSVC	0.3693333	0.3687021846068616
1990	NaiveBaye	0.3206666	0.320489
1990	LogisticRe	0.336	0.33611621334994995
1990	LinearSVC	0.3166666	0.31647004745011903
2010	NaiveBaye	0.3093333	0.3095575016776811
2010	LogisticRe	0.3166666	0.3168795509582507
2010	LinearSVC	0.2893333	0.2892955910921515

Figure 9: Within-Decade

Training_D	Testing_D	Classifier	Accuracy	Macro_F1
1970	1970	NaiveBaye	0.816	0.8158457308040552
1970	1970	LogisticRe	0.8286666	0.8285906129873366
1970	1970	LinearSVC	0.9773333	0.9773261324464355
1970	1990	NaiveBaye	0.2973333	0.29662648793587004
1970	1990	LogisticRe	0.3033333	0.3020900176762277
1970	1990	LinearSVC	0.2793333	0.27946639029550935
1970	2010	NaiveBaye	0.2886666	0.2870048436660168
1970	2010	LogisticRe	0.2846666	0.2823472927327003
1970	2010	LinearSVC	0.2793333	0.2781807182728192
1990	1970	NaiveBaye	0.328	0.32465782205952504
1990	1970	LogisticRe	0.3273333	0.32609040792003424
1990	1970	LinearSVC	0.3233333	0.32199967092651083
1990	1990	NaiveBaye	0.8286666	0.8288394212267545
1990	1990	LogisticRe	0.8293333	0.8293272788554802
1990	1990	LinearSVC	0.9673333	0.9673450086092303
1990	2010	NaiveBaye	0.314	0.31403034212630254
1990	2010	LogisticRe	0.3133333	0.31306895913889166
1990	2010	LinearSVC	0.296	0.29613982253843724
2010	1970	NaiveBaye	0.3186666	0.31694860918487816
2010	1970	LogisticRe	0.3286666	0.3264947318419498
2010	1970	LinearSVC	0.306	0.30480253361066084
2010	1990	NaiveBaye	0.318	0.318103
2010	1990	LogisticRe	0.3266666	0.32699001619903856
2010	1990	LinearSVC	0.2993333	0.29940887789070525
2010	2010	NaiveBaye	0.7953333	0.7958297497922431
2010	2010	LogisticRe	0.804	0.8041150935257532
2010	2010	LinearSVC	0.9653333	0.9653378882442316

Figure 10: Cross-Decade

Decade	Classifier	Accuracy	Macro F1
1970	Baseline	0.2	0.066667
1990	Baseline	0.2	0.066667
2010	Baseline	0.2	0.066667

Figure 11: Baseline

Classifier	With-In	CoCross	Cori	McNemar	p-value	Significance
NaiveBaye	439		24	370.1857	0	TRUE
LogisticRe	460		21	398.8441	0	TRUE
LinearSVC	516		2	508.0483	0	TRUE

Figure 12: McNemar’s test

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